Bachelor Project Applied Computer Science
(Academic year 2025—2026)

Federated Learning

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1 Scope & Vision

1.1 Background

The project was carried out at company Result d.o.o. It is a software development company, but also focus on data, artificial intelligence, and business solutions. They work on many projects, some of which are European. One such is eEarly — an open platform for analysis and gathering of health data [1]. It was also the umbrella for this bachelor work.

European Union has many regulations and laws related to data, and there are even more for each individual country, organisation,... This introduces challenges for projects like eEarly that utilize artificial intelligence and deal with sensitive medical data. The motivation for my bachelor project is to find the bridge between sensitive data, artificial intelligence, and a complex regulatory landscape.

The direction of the artificial intelligence model was disease prediction. The main sources of data were APNEA HRV+SPO2 DATASET [2] and a smart medical ring Wellue O2Ring [1] that a person can wear while it continuously performs medical measurements. The artificial intelligence model used heart rate and oxygen saturation (SpO2) to predict whether a person has sleep-related breathing disorder called obstructive sleep apnea (OBS) [3].

1.2 Problem statement

In an ideal situation, an artificial intelligence model could train on all data without restrictions. In reality, besides strict European Union laws, such as General Data Protection Regulation (GDPR) [4], [5], each hospital normally keeps its data isolated. This reveals the main issue: An artificial intelligence model can — without considerable legal and data privacy effort — only train on one hospital. Consequences of this are poor model performance with less accurate predictions, and the new technologies are not being utilized fully [6]. From this follows the proposal, which is not to loosen the law, but to explore techniques like federated learning [7] to satisfy both the regulations and the artificial intelligence's demand for data.

1.3 Intention

The main intention of the project is to develop a federated learning proof-of-concept (POC) for the company, which can later be used as a starting point for other projects and might be fully implemented, possibly in a healthcare environment. It might also become a new privacy-oriented artificial intelligence solution that the company could offer to clients for commercialization as a reusable asset.

1.4 Scope within the bachelor project (MVP – need to have)

The scope can be seen on the Table 1 on page 3. The requirement descriptions include terminology introduced in Section 3. "Must have" features describe the minimal viable product (MVP); without these, the project is not deemed finished, unless blockers are clearly stated.

1.5 Ultimate scope (want to have)

Ultimate scope of the project is defined with features of "Should have" and "Could have" importance in the Table 1 on page 3. One can see that as importance lowers, features become closer to those of realistic setting in which federated learning would have been deployed, and more unknown challenges might appear.

[1] https://getwellue.com/pages/o2ring-oxygen-monitor

2 Software Requirements Specification

2.1 Target group

The primary target group of the project is developers. Most important aspect to consider for this target group is the knowledge to read code and its documentation. Their needs are not only to be able to understand code and its documentation, but also know how to interpret the results at the most basic level, and most important, they need to know how to run the project.

For the umbrella project eEarly, there are more potential stakeholders such as patients, whose sensitive data is guaranteed to remain private during the artificial intelligence training process, and doctors that would receive improved disease predictions. Perhaps there are even stakeholders from the legal perspective since the sensitive data is not moved and there is less possibility to breach current legislations and less need to adapt it. Though that is out-of-scope for the federated learning project.

2.2 Method

In order to discover, track, and prioritize all requirements, I was, especially in the beginning, asking a variety of questions to my mentor at Result d.o.o. and kept sorting them by importance using the MoSCoW framework [8]. Besides being very simple and clear, this method has proven itself in my previous projects, which is the main reason why I chose it as my main tool at this stage. All of this was done on Jira in a kanban board, so that it has always remained organized and visible to me and mentors.

2.3 Requirements

Functional requirements can be seen in the Table 1. "Must have" requirements form the minimal viable product (MVP). Out of all requirements, I have created epics that can be seen in the list below. I have not used user stories, since the requirements are very broad with many tasks.

1. As a developer,
   I want to run a single-machine and non-containerized simulation of apnea predictions using federated learning,
   so that the whole structure is as simple as possible.

2. As a developer,
   I want to run a single-machine and containerized (with Docker) simulation of apnea predictions using federated learning,
   so that the whole project can be easily started by anyone on any machine.

3. As a developer,
   I want to run a multi-machine and containerized (with Docker) simulation of apnea predictions using federated learning,
   so that the structure comes closer to real and production-ready implementation.

4. As a developer,
   I want to run a multi-machine and containerized (on Kubernetes) simulation of apnea predictions using federated learning,
   so that the structure comes closer to real and production-ready implementation.

5. As a developer,
   I want to have a differential privacy mechanism implemented on top of federated structure,
   so that I can assure the system also secures weights and parameters that are sent to a central server.

Non-functional requirements:

- A developer needs to understand
  - the code,
  - the documentation,
  - how to interpret results at the most basic level, and
  - how to run the project.

Table 1: Functional requirements with their importance, higher rows correspond to higher priority.

Importance	**Requirement**
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning.
Should have	Single-machine and containerized (with Docker) simulation of apnea predictions using federated learning.
Could have	Differential privacy mechanism implemented.
Could have	Multi-machine and containerized (with Docker) simulation of apnea predictions using federated learning.
Could have	Multi-machine and containerized (on Kubernetes) simulation of apnea predictions using federated learning.
Will not have	Non-basic visualizations.
Will not have	Advanced artificial model.
Will not have	Other security features like secure multiparty computation.
Will not have	Production ready features, e.g. blockchain technologies like swarm learning, compatibility with NVIDIA FLARE.

3 Preliminary Research

3.1 Federated Learning

Imagine there are researchers working on a project and they do not tell each other anything about themselves, their names, education, age… They only contribute with their current knowledge without revealing their personal information. This is what federated learning tries to achieve [9]. Later on I talk about global and local models, in this example, researchers play a role of local models, and the project would be a global model.

Federated learning is a *decentralized learning* technique, which means the learning is not accomplished centrally on a single device — normally a server — but is kept on the edge, close to the main source of the data. Since it is decentralized, this also leads to great benefits regarding *privacy* and regulations as data is not moved out of its safe zone. In practice, we accomplish this using two types artificial intelligence models: global and local. The algorithm [10] contains the following steps:

1. Global model is initialized.
2. Global model sends a copy of himself to all the local models. In a medical example, local models would be hospitals.
3. Local models train on sensitive (medical) data. There is no data exchanged between *any* of the models.
4. Local models send the weights to the global model. One can image they share their knowledge, without data.
5. Global model combines the weights — usually by averaging them —, and becomes better at predicting diseases, for example.
6. The process repeats from the second step. Each local model now becomes better as it receives the average "knowledge" from all others.

3.1.1 Types

There are two main types of federated learning that can be implemented based on the similarities between datasets of clients. A definition of a dataset is essential: A *dataset* is a collection of data, commonly structured into a format comparable to a spreadsheet where each rows represent a record (also called example). A record consists of values that represent either a feature or a label. A feature is input for a machine learning model, such as heart rate. A label is either output of a machine learning model — in a disease prediction examples would be 0 ("Obstructive sleep apnea not detected") or 1 ("Obstructive sleep apnea detected") — or an answer for a machine learning model while it is being trained on so called labeled data [11].

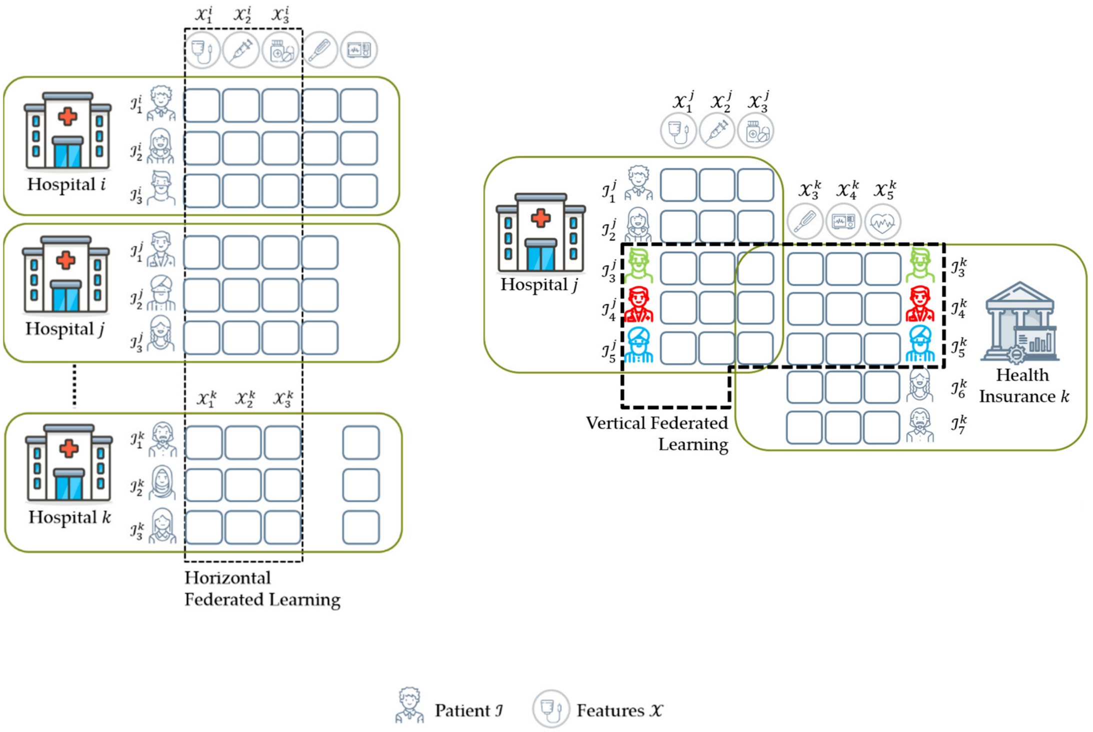
Horizontal Federated Learning In horizontal federated learning, clients' datasets describe same features meanwhile the data is different [5], [12]–[14]. For distinction with vertical federated learning, it is important here to keep in mind an almost obvious fact — By saying "the data is different" between clients, we mean that the feature that uniquely identifies a record, such as an email or a national identification number, is never duplicated across clients' datasets. For example, there are most likely many hospitals worldwide that have treated sleep apnea and have targeted same medical features, such as heart rate and oxygen saturation, but collected different data for different patients (see Figure 1). A federated learning model could predict sleep apnea based on all hospitals' client models.

Vertical Federated Learning In vertical federated learning, clients' datasets describe different data and different features, except for identification features such as identification number [5], [12]–[14]. For example, a country has a hospital for sleep disorders that collects features like patient national identification number, heart rate and oxygen saturation, and a cancer hospital that collects patient national identification number, x-ray scan images, blood information (visualized in Figure 1, although a health insurance is used instead of a cancer hospital) [15]. A federated learning model could connect the knowledge of both hospitals using patient identifier, find patterns, and become more holistic in its predictions.

Federated Transfer Learning In federated transfer learning, clients' datasets describe different data and different features, including the identification features [5], [12]–[14]. For example, two countries have two hospitals and there is very little overlap between features or data — the patients' identification is also mostly different — yet federated transfer learning can be applied in this scenario to combine their knowledge.

We will focus on horizontal federated learning prototype, because our features remain same across clients, namely hospitals.

Figure 1: Visualization of main types of federated learning in healthcare scenarios, sourced from [5]. On the left horizontal federated learning from different data with same features. On the right vertical federated learning from different data and mostly different features.



3.2 Related projects

Numerous similar projects with alike problems can be found, though there are some more noticeable and important.

In 2017, Google introduced federated learning while implementing the "Next word suggestion" feature on their Gboard. They took user interaction as their input and trained an artificial model to predict users next word. Without this innovating project, all the private typing data from millions of devices would have to be transferred to a central server. At this point, we can easily understand Google's main concerns of standard data pipeline: Bandwidth usage and latency [9], [16].

During COVID-19 pandemic, researchers all over the world have established connections and collaborated to create an artificial intelligence platform that predicts whether you have the disease based on CT (X-ray) chest scans. This was an initiative called Unified CT-COVID AI Diagnostic Initiative (UCADI), and the reason I am mentioning it here is that they have made it all possible by using federated learning [16], [17].

3.3 Related solutions

Main characteristics of federative learning described in Section 3.1 are decentralized learning and privacy; I use this two for finding related solutions below.

3.3.1 Decentralized Learning Techniques

Fog computing is an improved approach on top of federated learning. It originates from the network challenges of federated learning, focuses on layered architecture of communication between local and global model, and aims to reduce network traffic [18].

Being a decentralized technique, federated learning still has a central, global model that aggregates all parameter updates. *Gossip learning* — also called distributed federated learning (DFL) [19] — is even simpler in this sense, as it does not have a global model. The updates and aggregations are all taken care by the local models themselves [20].

Similar to gossip learning is *swarm learning*. The main difference is its use of blockchain peer-to-peer communication that is configured between the local models. This enables swarm learning to be a very related solution, with even better potential privacy and fault tolerance (because there is no single point of failure like the global model server) [19].

Centralized learning does not come into considerations because the movement of data to a central server is restricted by data regulations. Federated learning is perfect for the MVP, as it is simple and the core concept for the decentralization alternatives above that introduce unnecessary complexity for the project at this stage, although techniques like swarm learning would be necessary on a production-ready.

3.3.2 Privacy Preserving Techniques

In general, *homomorphic encryption* creates a possibility to make operations on encrypted data, without ever decrypting it. This can be applied both to normal and federated machine learning, although it is rather an addon than a replacement for the latter. The challenges include computational overhead and lack of support for some operations, thus finding a balance between privacy and performance is an important question [21], [22].

Secure multiparty computation allows to split private data from many participants, share it in this split form, and let any participant compute (public) functions, such as average and maximum without knowing any other participants' private data [23]. In federated learning, sharing weights with the central server still brings up some privacy concerns that can be handled by applying secure multiparty computation [24]. This adds some computation cost, along with even more communication overhead [21].

Differential privacy introduces random noise to private data while still maintaining its statistical properties [25]. This means that individual values like age are uninterpretable for any middleman between global and local models, thus guarding the system from inference and poisoning attacks. For example, inference attack would mean that a middleman can infer whether a person has a certain disease from the outputs of the global model [26]; data poisoning attack would mean that a middleman intercepts and modifies the weights sent to the global or local model, making the predictions of the model less accurate or corrupted [27].

All privacy alternatives mentioned above are strongly related, but optional for the scope of this work. They clearly point out main weak parts of the federated learning which have to be kept in mind in phases later on.

3.4 Required technology

There were minimal hardware requirements to develop the most important parts of the project. Must have requirements were satisfied only by using a single computer, and a dedicated remote server with a strong computing capabilities, specifically meant for faster model training, although it was not a necessity. Later on, another computer was added to configure the federated network in a multi-machine environment, making it more realistic. Ring sensor was not used, because other requirements were prioritized.

Software requirements consisted mostly of a programming language, integrated development environment and a virtualization technology. Since the apnea prediction model was already coded in a programming language Python (with Tensorflow machine learning library), I made sure this was also the case for the federated learning network. Federated Flower network was coded in Python and specifically configured for use with Tensorflow. Coding itself was done in IntelliJ and Visual Studio Code; the latter was used for easier SSH access to the remote machine learning server mentioned in the previous paragraph. Virtualization technology used after developing the minimal viable product was Docker, which made it possible to run all necessary code on separate computers without any other software. No other programming language or software was used or tested during this project.

3.5 Comparative study

From [28]–[30], I have extracted four main open-source frameworks suitable for the project. One of them, namely FedML, had vastly different opinions among papers, so I did not take this framework into consideration.

Flower: It's main characteristic is easy integration with different deep learning frameworks, and that it offers a beginner- and prototype-friendly entrance into federated learning [29], [31]–[33]. As Flower was new in 2025, research papers from that year showed it has lacked community support and documentation [29], [31], which has changed the next year, and is now apparently considered to have strong documentation [32]. It is especially well-suited for edge IoT devices and production ready [31], although additional privacy features have to be engineered manually [32], unlike with frameworks like NVIDIA FLARE.

NVIDIA Federated Learning Application Runtime Environment (NVIDIA FLARE): It's main characteristic is enterprise-readiness [31]–[34], with many built-in features, especially for privacy [34]. It is overall very good in all aspects (also documentation) [28], [31], which is the reason it might seem heavy for smaller projects [32]. The main disadvantages is that you are restricted to the NVIDIA ecosystem [31].

TensorFlow Federated (TFF): It's main characteristic is that is most widely used, but complex, only compatible with TensorFlow, and can be integrated into the least environments [29], [31].

The choice comes down to Flower or NVIDIA FLARE. Luckily, these two have recently announced collaboration and compatibility between each other [35]–[37], meaning Flower can be integrated into the NVIDIA runtime environment and immediately leveraging enterprise-ready environment from NVIDIA and simplicity from Flower. For this reason, Flower will be used to create the prototype.

3.6 Final selected technology / project tools

Which project tools did you use?

Which development methodologies did you apply?

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3.7 Choice reflection

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4 Reflection

4.1 General Reflection

4.1.1 Achieved objectives

Were the project objectives achieved?

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4.1.2 Test framework / technique

Which test framework or technique was chosen and why?

Which parts of the project were or were not tested, and why?

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4.1.3 Security

Was security considered in the project and how was it addressed?

Where do you see possible improvements in the project's security?

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4.1.4 Improvements

Where do you see improvements in general?

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4.1.5 Added value of the project

What do you consider the greatest added value of the project?

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4.1.6 Future opportunities

What are the next steps or future opportunities for the project?

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4.1.7 Next steps

What are the next steps or future opportunities for the project?

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4.2 Personal Reflection

If you were to do the project again, what would you approach differently?

What impact do you think the research or developed tool has on people (ethics/inclusion)?

What impact do you think the research or developed tool has on society (economy/policy/education)?

What impact do you think the research or developed tool has on the climate (energy/sustainability)?

What will stay with you most about the project?

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4.3 Feedback

What did you think overall about the integrated bachelor project and the distribution with the traineeship?

Do you have suggestions for improvement for the integrated bachelor project?

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5 Appendices

Meeting minutes van vergaderingen

The backlog

The story mapping (with different releases)

Example mappings (if available or in another format)

Gherkin scenarios (if available)

Login credentials

Installation manual

User manual

5.1 Meeting minutes

5.2 Backlog

5.3 Story mapping

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